CONTACT OR CONTOUR? SUPERVISING A BOUNDARY MATTERS MORE THAN WHICH BOUNDARY FOR SEPARATING TOUCHING NUCLEI

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ABSTRACT

Supervising the borders where nuclei touch, as in the winning 2018 Data Science Bowl solution, is a popular way to keep touching nuclei from merging, but its effect is rarely separated from the design choices that change with it. We measure it in matched ten-seed experiments with image-paired confidence intervals on BBBC039, two NucleusSegData test sets (Huh7, HepG2), and 53 external DSB2018 images, against full-contour, semantic-watershed, and scratch-trained Cellpose baselines. In an exploratory comparison, adding a boundary head, with marker-based decoding, to the watershed baseline raised aggregated Jaccard index (AJI) by 0.027–0.041 on BBBC039 and 0.035–0.036 externally, and object F1, the pre-specified endpoint, by 0.023–0.028 and 0.028–0.040. Which boundary was supervised mattered less: on BBBC039 a contact-only target beat the full-contour target by 0.014 AJI, robust to seed and image resampling and to test-time augmentation, with fewer merged touching nuclei (21% vs. 27%), but by only 0.005 in object F1. Elsewhere the contact advantage was small or conditional: externally the targets did not differ detectably in AJI, and contact's 0.011 object-F1 gain did not survive seed resampling; Huh7 excluded a gain of the BBBC039 size; HepG2 gained only after re-tuned post-processing; and a contact head lowered Cellpose's AJI. Scratch-trained Cellpose, which has no boundary head, had the best panoptic quality on every test set. For foreground-based networks on these fluorescence benchmarks, adding a boundary head with its decoder is worth doing; which boundary is a secondary choice to confirm on held-out data.

***Index Terms*—** nucleus instance segmentation, touching nuclei, boundary supervision, Cellpose, empirical comparison

1. INTRODUCTION

Covering two touching nuclei with one object barely changes foreground Dice yet corrupts cell counts and per-cell measurements; on BBBC039, 29% of nuclei touch another. Instance methods separate nuclei with contour classes [1, 2], distance or shape representations [3, 4, 5], gradient flows [6], or multi-stage attention [7]. A simpler, widely used heuristic is to supervise where nuclei touch: U-Net up-weighted the gaps between touching cells [8], a multiclass loss added a touching-region class [9], and the winning 2018 Data Science Bowl solution predicted touching borders together with masks [10, 11].

New methods typically change backbone, decoder, targets, and training recipe together, and single-run comparisons can mistake seed variance for method effects [12]. We ask how much contact targets help relative to the choices that change with them, separating matched from composite contrasts. Our contributions are: (1) a ranking of design effects (Fig. 3, AJI): where both were measured, adding a boundary head raised AJI by 0.027–0.041 and object F1 by 0.023–0.040, while the boundary target changed AJI by at most 0.014 on any of four test sets, and not consistently; (2) a multi-seed, image-paired test of contact versus full-contour targets with identical network, loss, selection, and decoder, robust in AJI to seed and image resampling on BBBC039; and (3) an error analysis of where contact targets help (fewer merged touching nuclei on BBBC039) and where they do not (Huh7, inside Cellpose).

2. METHODS

Boundary targets. Let L be the reference instance map (0: background), $N_k(x)$ the k-neighbourhood of pixel x, and $\mathcal{D}$ dilation by a radius-1 disk. The targets differ mainly in whether the neighbour that defines a boundary may be background (and in connectivity):

$$C = \{x : L(x) > 0,\ \exists x' \in N_8(x) : L(x') \notin \{0, L(x)\}\},$$
$$B = \{x : L(x) > 0,\ \exists x' \in N_4(x) : L(x') \neq L(x)\}, \tag{1}$$

with contact target $c = \mathcal{D}(C) \cap \{L > 0\}$ and full-contour target $b = \mathcal{D}(B) \cap \{L > 0\}$ [1, 2]. Thus c keeps only borders between nuclei, the cue that decides separation, whereas b also marks nucleus–background edges that the foreground head already resolves; on BBBC039, c covers 0.34% of pixels and b 4.9%. Targets are built at native resolution and re-

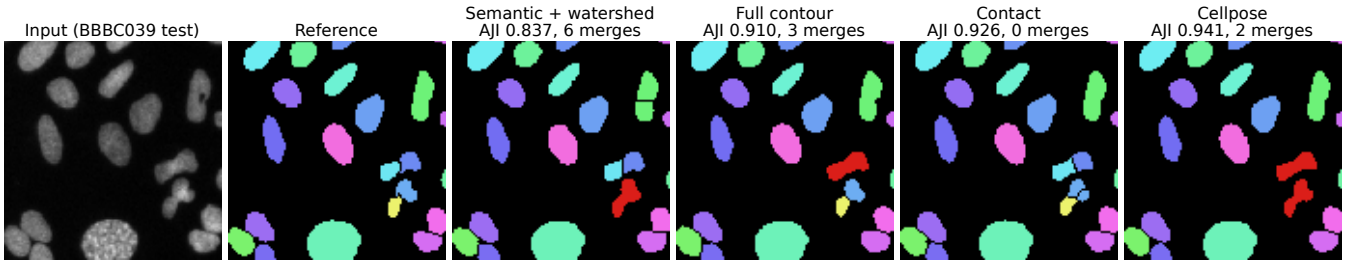


Fig. 1. BBBC039 test crop chosen by a fixed rule (contact model's upper-median seed by object F1, its median image, the window with most contacts; same seed for all models). Objects take the colour of the reference nucleus they mostly cover; red: merging objects; grey: unmatched. Titles: whole-image AJI, merging objects.

sized (nearest neighbour) with the image; the foreground target $y_f = \{L > 0\}$ is left intact.

Network and loss. A UNet++ [13] with an ImageNet-pretrained EfficientNet-B7 encoder [14] (decoder widths 256–16) ends in one two-channel layer whose sigmoids give foreground p_f and boundary p_c, trained on c or, for the control, on b. The loss is

$$\mathcal{L} = \tfrac{1}{2}[\mathcal{L}_D(p_f, y_f) + \mathcal{L}_B(p_f, y_f)] + \tfrac{1}{4}[\mathcal{L}_D(p_c, c) + \mathcal{L}_{B,w}(p_c, c)], \quad (2)$$

with soft Dice $\mathcal{L}_D$ and binary cross-entropy $\mathcal{L}_B$; $\mathcal{L}_{B,w}$ weights positive pixels by $w = \min(50, \max(1, (1-r)/r))$ for batch target ratio r, which sits at the cap for c and near 20 for b, so the target contrast includes this weighting difference.

Decoding. A boundary prediction may split a foreground component but never joins or deletes one: (i) $F = \{p_f > .5\}$ after filling holes and removing objects smaller than s pixels ($s = 16$; NucleusSegData: chosen on training images); (ii) markers are the 4-connected components of $F \setminus \{p_c > \tau\}$ with at least s pixels, $\tau \in \{.3, \ldots, .7\}$ chosen on validation (NucleusSegData: $\{.1, \ldots, .7\}$, training images); (iii) within each 4-connected component of F, every pixel takes its Euclidean-nearest marker in that component (component-preserving), or the component is flooded from its markers by watershed on p_c; (iv) a component without markers stays one object. Given the foreground, merges thus stem mainly from missed boundaries and splits from spurious ones. An earlier global nearest-marker decoder, found faulty after test scoring, handed marker-less components to neighbours and more than doubled the contact–contour AJI gap (AJI .870 vs. .838); no reported result uses it.

Matched controls and baselines. With identical network, loss, selection, and decoder, we compare c with b, and, with a modified foreground reconstruction, with a *carved* target that, like one variant of the 2018 winning solution, removes contact pixels from y_f (decoded with $F = \{p_f > .5\} \cup \{p_c > \tau\}$). A one-channel *semantic* network is decoded by connected components (CC) or distance-transform watershed (h-maxima on a smoothed distance map, h, σ chosen on validation); its checkpoints are selected by validation object F1 after watershed, as for the two-head models. Cellpose [6] (v3.1.1.2) is trained from scratch on the same split (its released models saw Data Science Bowl images, which share BBBC039's source) with its default recipe (500 epochs; inference diameter: median square-root nucleus area, 25/32 pixels; Cellpose's convention, 28/36, changed mean AJI by $\leq$0.004); flow and probability thresholds are chosen on validation. We add contact supervision to it as a fourth output trained with the contact loss above (decoded normally or by splitting objects along predicted contacts), and as weights $1 + Wc$ on Cellpose's flow and probability losses, screened on validation ($W \in \{3, 10\}$) by a pre-set rule (object F1 $+0.002$, no seed declining).

Error taxonomy. A reference nucleus is *merged* if its majority lies in a predicted object that also holds another nucleus's majority, *split* if it holds the majority of two or more objects, and *unmatched* without an IoU$>$0.5 match; strata (touching another nucleus; small, $<$128 native pixels) were fixed before analysis. Rates pool nuclei per seed, then average over seeds.

3. EXPERIMENTAL SETUP

Data. BBBC039v1 has 200 520×696 Hoechst images of U2OS cells [15, 11]; we use the official 100/50/50 split. Expanded NucleusSegData [7, 16] has 25 training, 11 Huh7, and 25 HepG2 test images. For zero-shot testing we use the 53 grey-scale fluorescence images of the 2018 Data Science Bowl stage-1 test set [11], none duplicating a BBBC039 image (template-matching correlation $\leq$0.58).

Protocol. Base models resize percentile-normalized images to 512^2 and train for 20 epochs (AdamW, learning rate 3×10^{-4}, cosine decay, flips and rotations); checkpoints and τ are chosen on the 50 validation images by object F1 before test scoring. Base two-head rows re-decode these locked choices with the component-preserving decoder; rows with eight-view dihedral test-time augmentation (TTA) re-elect decoder and τ on validation from one grid shared by

Table 1. BBBC039 official test set, mean (seed SD) over ten seeds unless marked ([§]five seeds). Two-head rows use the component-preserving decoder (final setting: it or watershed on p_c, chosen on validation); [¶]200 epochs, intensity jitter, chosen on validation. Literature: published values, same split; [‡]scored on expanded-interior targets.

Model	DICE1	AJI	PQ	Obj.-F1
Foreground-based, base protocol (512^2)				
Semantic + CC	.9653(.0013)	.6779(.0133)	.7406(.0080)	.7520(.0084)
Semantic + watershed	.9653(.0013)	.8303(.0050)	.8371(.0029)	.8583(.0031)
Full-contour target	.9637(.0016)	.8568(.0042)	.8541(.0044)	.8817(.0044)
Carved foreground	.9650(.0025)	.8692(.0036)	.8619(.0039)	.8865(.0033)
Contact target	.9660(.0023)	.8712(.0057)	.8610(.0047)	.8865(.0040)
Final setting (TTA, validation-selected decoder)				
Full contour, 512^2	.9676(.0006)	.8682(.0033)	.8664(.0025)	.8934(.0027)
Contact, 512^2	.9692(.0010)	.8792(.0051)	.8713(.0029)	.8959(.0030)
Contact, native resolution	.9715(.0008)	.8895(.0040)	.8807(.0028)	.9049(.0023)
+ searched recipe[¶]	**.9721**(.0003)	.8965(.0023)	.8835(.0016)	.9069(.0013)
Flow-based (Cellpose 3, from scratch)				
Cellpose	.9702(.0016)	**.9053**(.0035)	**.8884**(.0036)	.9119(.0031)
+ contact head[§]	.9700(.0004)	.8975(.0013)	.8872(.0009)	**.9121**(.0011)
ASW-Net[‡] [19]	.9645	.9020	.8665	–
BISSG [20]	.9482	.8680	.8629	–
Caicedo U-Net [2]	–	–	–	.8980

both targets; all re-decodings are reported. Each arm was a separate locked study with these settings; a Gabor-encoder contact variant scored lower (AJI -0.009) and is not analysed. A final setting trains at native resolution (random 512^2 crops, 40 epochs) with TTA; a validation-only search (six recipes, two seeds) then chose 200 epochs with intensity jitter. NucleusSegData has no validation set: models train for 60 epochs (Cellpose: 500), keep the final epoch, and choose post-processing on training images.

Endpoints and statistics. Foreground Dice (DICE1), AJI [17], panoptic quality (PQ) [18], and object F1 over IoU 0.50:0.05:0.90 [2], as image means; NucleusSegData uses the Bilkent nucleus F1 [16] (one-to-one matches covering half of each other, pooled per subset). The locked protocols designated object F1 as primary; AJI, a main endpoint of prior BBBC039 work [19, 20], was emphasized after the locked runs and is reported alongside it. Contact vs. contour (base protocol) was the pre-specified primary contrast, but as earlier BBBC039 test results had been seen and the decoder was fixed after test scoring, all intervals are exploratory and unadjusted. Count error is the mean absolute per-image count difference. Seeds are averaged within each test image; 95% percentile CIs use 10,000 image-paired bootstrap resamples, conditional on the trained models; a two-way bootstrap that also resamples each arm's seeds checks retraining variability.

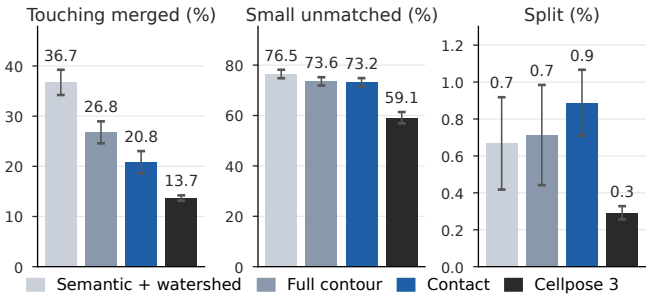


Fig. 2. Error profile on BBBC039 test (foreground-based models: base protocol; mean and seed SD over ten seeds) as a percentage of the 1,650 touching, 509 small (<128 pixels), and 5,720 reference nuclei.

4. RESULTS

4.1. Supervising a boundary mattered more than which one

Supervising contacts instead of contours added 0.014 AJI ($[.011, .018]$; 46/50 images) and object F1 $+0.005$ ($[.002, .008]$); with TTA (same decoder grid and selection rule for both), 0.011 ($[.008, .014]$; 45/50; object F1 $+0.003$, $[-.0001, .005]$). Resampling seeds as well as images kept both AJI intervals above zero ($[.009, .020]$, $[.006, .016]$), but not the object F1 interval with TTA ($[-.001, .006]$; base $[.0002, .010]$). Adding a boundary head[†] to the watershed baseline mattered more (exploratory; the decoder changes with it): AJI rose by 0.041 ($[.032, .051]$; 48/50) with contact and 0.027 ($[.019, .035]$) with contour, 0.012 ($[.004, .021]$) more than the target effect, and object F1 by 0.028 ($[.022, .035]$) and 0.023 ($[.017, .030]$), 0.019 ($[.012, .026]$) more; all stayed above zero with seeds also resampled (differences $[.001, .023]$, $[.009, .028]$). Carving contacts out of the foreground lowered AJI by 0.002 ($[.0004, .0036]$), and a validation-searched recipe raised it by 0.007 ($[.004, .010]$). Cellpose exceeded our best contact model by 0.009 ($[.004, .013]$; two-way $[.004, .014]$).

4.2. On BBBC039, contact targets merged fewer nuclei

Contact supervision was accompanied by (Figs. 1, 2) fewer merges of touching nuclei, from 36.7% (watershed) and 26.8% (full contour) to 20.8% (contact minus contour -5.9 points, $[-7.7, -4.4]$, seed and image resampling $[-8.6, -3.5]$; 44/50 images), and of all nuclei, from 10.8% and 8.2% to 6.3%, while splits stayed below 1%, although splits of touching nuclei rose from 1.3% (contour) to 2.3%. Per-image count error fell from 10.1 (watershed) and 7.2 (contour) to 6.6 nuclei. Unmatched small nuclei, by contrast, were similar across foreground-based models (73–76%); the native-resolution recipe lowered this to 64% for the contact

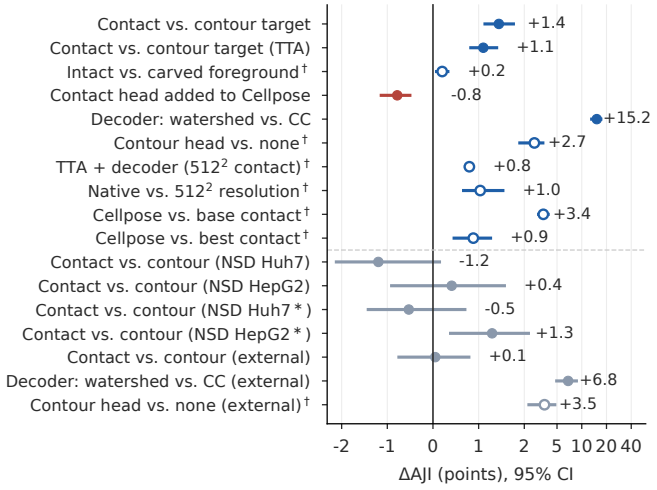


Fig. 3. Effect of each design choice on AJI (points, AJI×100; mean of image-paired differences, 95% bootstrap CI; symmetric-log axis). Rows above the dashed line use BBBC039 test; below it, NucleusSegData (NSD; *widened grid and TTA for both targets) and the external DSB2018 images. Open markers (†): composite comparisons that also change decoder, reconstruction, recipe, or resolution.

model (no TTA; post hoc), so most small nuclei stayed unmatched.

4.3. A contact head did not help Cellpose

Cellpose merged fewer touching nuclei and left fewer small nuclei unmatched (Fig. 2). In the tested recipe, the auxiliary contact head (five seeds) lowered AJI relative to the ten baseline seeds by 0.008 ($[.005, .012]$; worse on 39/50 images) without materially changing PQ or object F1 (-0.001, $+0.000$; equal seeds, resampled: AJI -0.007, $[-.011, -.002]$); validation never selected contact-based splitting in any seed. The contact-weighted loss failed its validation screen and was not tested.

4.4. Second benchmark and external images

On expanded NucleusSegData (Table 2), the contact model matched the published multi-stage AttentionBoost in nucleus F1 and was higher with a widened training-selected post-processing grid and TTA. In the base setting the targets did not differ detectably (Huh7 AJI -0.012, $[-.024, .002]$, 2/11 images; HepG2 $+0.004$, $[-.009, .016]$); the Huh7 interval excludes the BBBC039 gain of 0.014. Re-decoding both targets with the widened grid and TTA (Table 2, *) kept Huh7's direction (-0.005, $[-.015, .007]$) but favoured contact on HepG2 ($+0.013$, $[.004, .023]$; 16/25; nucleus F1 $+0.005$). Scratch-trained Cellpose was 1.1 and 1.4 points above the best foreground-based model and had the best

Table 2. Expanded NucleusSegData, Bilkent nucleus F1; ours: mean (seed SD) over ten seeds, component-preserving decoder; AttentionBoost: five published runs [7]. *Widened training-selected grid and TTA. §Five seeds.

Method	Huh7	HepG2
Multi-task [7]	.934(.009)	.869(.005)
AttentionBoost [7]	.954(.005)	.919(.008)
Full contour	.9541(.0062)	.9162(.0079)
Contact	.9533(.0057)	.9170(.0080)
Full contour*	.9609(.0039)	.9240(.0029)
Contact*	.9575(.0051)	.9286(.0069)
Cellpose 3, scratch§	**.9715**(.0038)	**.9426**(.0038)

Table 3. Zero-shot results on 53 external DSB2018 fluorescence images (BBBC039-trained base models, ten seeds; count error in nuclei per image).

Model	DICE1	AJI	PQ	Obj.-F1	Count err.
Semantic + watershed	.803	.585	.555	.493	12.2
Full-contour target	**.812**	.620	.581	.521	**9.1**
Contact target	.811	.621	.589	.532	9.3
Cellpose	.796	**.633**	**.633**	**.584**	13.7

PQ (.870 and .795 vs. at most .859 and .785). Applied unchanged to the 53 external images (Table 3), a boundary head again improved on the semantic watershed in AJI (contour $+0.035$, $[.022, .049]$; contact $+0.036$) and object F1 ($+0.028$, $+0.040$). The targets did not differ detectably in AJI ($+0.001$, $[-.008, .008]$) or PQ ($+0.008$, $[-.0001, .016]$); contact had higher object F1 ($+0.011$, $[.002, .021]$; 35/53) but merged more touching nuclei (30.9% vs. 27.6%; $+3.3$ points, $[1.3, 4.8]$); neither difference was robust to seed resampling ($[-.008, .031]$, $[-0.4, 7.3]$). Cellpose exceeded the contact model in PQ ($+0.044$, $[.016, .072]$) but not detectably in AJI ($+0.012$, $[-.016, .039]$), with a larger count error ($+4.4$ nuclei per image, $[2.4, 6.9]$) but fewer unmatched predicted objects (12% vs. 24%).

5. DISCUSSION AND CONCLUSION

Where both were measured, adding a boundary head with its decoder helped more than the choice of boundary, in AJI and in the pre-specified object F1. Contact beat contour robustly only in BBBC039 AJI; elsewhere its advantage was conditional or absent. Cellpose, without a boundary head, had the best PQ on every test set, so boundary supervision is a lever for foreground-based networks, not a requirement. In practice: add a boundary head, never reassign marker-less components, compare with a flow-based model, and confirm the target on held-out data. Limitations: fluorescence images only, one annotation each; targets differ in positive weight and connectivity; base models trail a published U-Net (object F1 .887 vs. .898); no contour arm at native resolution; one Cellpose recipe; no StarDist [4] or foundation models [21, 22].

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using open-access images of cultured cell lines (BBBC039 [15], NucleusSegData [7], DSB2018 [11]). It involved no human subjects or animals, and ethical approval was not required.

7. ACKNOWLEDGMENTS

This work was supported by U.S. Department of Energy grant DE-SC0025403. The authors declare no competing interests. OpenAI Codex and Anthropic Claude assisted with analysis scripts, statistics, figures, and language editing of all sections; the authors designed and verified the experiments and take responsibility for the manuscript. Code and per-image results: `https://github.com/vkd14/contact-or-contour`.

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